

eWater Monitor

REST API & MCP Server Reference

Base URL: `/api` (REST) · `/api/mcp` (MCP)

Auth: eWater credentials via `POST /api/ewater/credentials`

MCP auth: `Authorization: Bearer <MCP_BEARER_TOKEN>`

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All REST endpoints are served under the base path `/api` and communicate in JSON. Most endpoints require eWater credentials to be configured first (see Credentials below). Unauthenticated requests receive `401 { "error": "No credentials configured" }`. Upstream eWater API errors are proxied back as `502 { "error": "eWater API error: ..." }`.

Credentials

GET `/api/ewater/credentials`

Check credential status

Returns whether eWater credentials are configured and when the session token expires.

```
{ "isConfigured": true, "environment": "live", "tokenExpiresAt": "2026-08-04T16:00:00Z" }
```

POST `/api/ewater/credentials`

Save eWater credentials

Stores credentials and validates them by obtaining a session token immediately. Returns `401` when credentials are invalid.

Body: `{ "username": "user@example.com", "password": "..." }`

DELETE `/api/ewater/credentials`

Clear saved credentials

Removes stored credentials and invalidates the session. Subsequent API calls return `401`.

Assets

GET `/api/ewater/assets`

List all assets

Returns every asset (community taps, controllers, eSENSE sensors) visible to this account.

Fields: `id`, `name`, `type`, `status`, `isOnline`, `batteryVoltage`, `waterSystemName`, `countryName`.

GET `/api/ewater/assets/:assetId`

Get asset details

Full detail for one asset: basic info, health status, connectivity, and power — fetched in parallel from the eWater State and Query APIs.

Name	In / Type	Required	Description
<code>assetId</code>	path · string	required	Numeric asset ID

GET `/api/ewater/assets/:assetId/tech`

Comprehensive technical status bundle

Aggregates all technical data for one asset in a single call (12 upstream requests in parallel): connectivity, power, flow, usage, status values, firmware, identifiers, recent commands, EWC settings, and entity hierarchy for water system / country resolution. Primary endpoint used by the asset detail page.

Key fields: `lastCommsDt` , `batteryVoltage` / `batteryTrend` / `batteryTodayHigh` / `batteryTodayLow` , `flowRateHour` / `flowRateToday` / `flowRateWeek` , `litresDispensedToday` , `imeis[]` , `firmware[]` , `healthFlags` , `tamperSwitchState` , `ewcFcf` / `ewcLcf` / `ewcFx` / `ewcPreload` / `priceOfWater` .

GET `/api/ewater/assets/:assetId/ewc`

EWC calibration settings

Returns the Electronic Water Controller device configuration: FCF, LCF (ticks per litre — the calibration factor), currency/FX, preload, price per litre, and all other EWC parameters.

GET `/api/ewater/assets/:assetId/esense-charts`

eSENSE time-series charts

Chart data for eSENSE tank-level assets over a configurable window.

Name	In / Type	Required	Description
<code>assetId</code>	path · string	required	Numeric asset ID
<code>days</code>	query · integer	optional (default 3)	Days to look back

Response includes: `tankHeight[]` (water & chlorine depth with min/max bands), `dailyInflow[]` (litres per day), `voltageHistory[]` , `voltageStatus` , `flowRateHistory[]` (per-dispense L/min), `dispenseVolumes` (histogram + KDE curve + suggested LCF calibration model).

GET `/api/ewater/assets/:assetId/meter-reading`

Tick accumulator meter reading

Extracts the lifetime tick accumulator from the most recent `HEALTH_STATE` (0x19) EWC packet in the last 14 days and converts it to litres using the asset's LCF.

Response: `{ ticks, lcf, litres, timestamp, found }`. `found: false` = no `HEALTH_STATE` packet received recently.

GET `/api/ewater/assets/:assetId/flow-rate`

Most recent flow rate

Most recent valid dispense flow rate (L/min) from EWC logs in the last 24 hours.

Response: `{ flowRate, timestamp, timedOut }`. `timedOut: true` = no dispense in 24 h.

GET `/api/ewater/assets/:assetId/telemetry`

Raw telemetry log

Last 50 raw telemetry entries for an asset (last 7 days), newest first. For fully decoded entries use `/logs`.

GET `/api/ewater/assets/:assetId/logs`

Decoded protocol logs (cursor paginated)

Decoded asset communication logs, newest first. EWC 2.5 and Shengda NB-IoT frames are decoded server-side. Shengda frames include a `shengda.description` human-readable summary.

Name	In / Type	Required	Description
<code>assetId</code>	path · string	required	Numeric asset ID
<code>before</code>	query · ISO timestamp	optional	Return only entries before this time — use last entry's timestamp to page
<code>protocol</code>	query · string	optional	Filter: Ewc2_5 CommandApi_1 4CCv1 Gadwall
<code>limit</code>	query · integer 1-100	optional (default 50)	Max entries per call
<code>windowDays</code>	query · integer 1-30	optional (default 7)	How many days back to scan

Response: `{ entries[], nextBefore, hasMore }`. Pass `nextBefore` as `before` to page forward.

GET `/api/ewater/assets/:assetId/packets`

Raw NB-IoT packet logs

Decoded Shengda NB-IoT CBOR/LwM2M packet logs, merged across all IMEIs registered to the asset (including secondary devices not in the identifier registry).

Name	In / Type	Required	Description
<code>assetId</code>	path · string	required	Numeric asset ID
<code>hours</code>	query · integer 1-72	optional (default 24)	How many hours back
<code>limit</code>	query · integer 1-100	optional (default 50)	Max packets
<code>imei</code>	query · string	optional	Restrict to a single IMEI

Tags & Households

GET `/api/ewater/tags`

List registered NFC tags for a water system

Paginated list of NFC tag IDs that have completed registration in the given water system.

Name	In / Type	Required	Description
<code>waterSystemId</code>	query · integer	required	Water system ID from <code>/ewater/entities</code>
<code>offset</code>	query · integer	optional (default 0)	Pagination offset
<code>limit</code>	query · integer 1-500	optional (default 100)	Max tag IDs to return

Response: pagination envelope with `tagIds[]` . `totalCount` = full registered fleet size.

GET `/api/ewater/tags/:nfcId`

Get tag info and household

Registration details for an NFC tag plus the linked household record.

Name	In / Type	Required	Description
<code>nfcId</code>	path · string	required	8-character hex tag ID (e.g. A7370000) — case-insensitive

Response: `{ tag: { primaryAssetId, householdId, creditBalance, disbursementCount, topUpCount, signUpDate, lastUsageDate, deleted, blacklisted }, household }` .

GET /api/ewater/tags/:nfcId/usage

Tag dispense event history

Individual dispense events for an NFC tag from the EWC packet log of the tag's primary asset.

Name	In / Type	Required	Description
nfcId	path · string	required	NFC tag ID
days	query · integer 1-90	optional (default 30)	Days back to scan
offset	query · integer	optional (default 0)	Pagination offset
limit	query · integer 1-200	optional (default 100)	Max events

Event fields: timestamp , litres , creditConsumed , eventType (Dispense | No Credit).

GET /api/ewater/households/:householdId

Get household

Household name and registration details by UUID (from tag.householdId). Phone number and address are omitted per eWater data-protection policy.

Disbursements (Usage API)

Per-day aggregated dispense totals from the eWater Usage API. Each day bucket includes: date , estimateTotalLitres , totalTicks , totalSeconds , totalCredits (null = no-credit event), readingCount . Period-level summaries totalLitres and totalReadings are also returned at the top level.

GET /api/ewater/tags/:nfcId/disbursements

Tag disbursements

Per-day totals for one NFC tag. If assetId is supplied, scoped to that tap only; otherwise aggregated across all taps.

Name	In / Type	Required	Description
nfcId	path · string	required	NFC tag ID
days	query · integer 1-365	optional (default 30)	Days to look back
assetId	query · string	optional	Scope to a specific asset

GET /api/ewater/assets/:assetId/disbursements

Asset disbursements

Per-day totals from one tap across all NFC tags.

Name	In / Type	Required	Description
assetId	path · string	required	Numeric asset ID
days	query · integer 1-365	optional (default 30)	Days to look back

Calibration

POST /api/ewater/assets/:assetId/detect-sensor-range

Auto-detect eSENSE sensor range

Correlates recent VSEN ADC readings from EWC packets with the eWater tank-height API to back-calculate the sensor's full-scale range in metres. Persists the result for use by alert rules.

Response: { sensorRangeMetres1, sensorRangeMetres2, sensorRangeMetres3, vsen1, vsen2, vsen3, depthMetres1, timeDeltaSeconds }. Returns 422 when no sensor data is available.

POST /api/ewater/assets/:assetId/apply-calibration

Write LCF + Preload to device

Sends RequestSettingChange to the eWater Command API for both LitresConversion and Preload. The device applies changes on its next comms cycle.

Body: { "lcf": 512, "preload": 0 } — both integers.

Response: { success, results: [{ settingKey, success, error }] }.

POST /api/ewater/assets/:assetId/reset-meter

Reset tick accumulator

Sends ResetTickAccumulator to the eWater Command API. IMEI is resolved automatically.

Body: { "litres": 15420.5 } **Response:** { litres, success, error }.

Dashboard & Utilities

GET `/api/ewater/dashboard`

System health dashboard

Aggregated health snapshot: online/offline/fault counts, power and flow fault tallies, recent system-level alerts.

Name	In / Type	Required	Description
<code>lifecycleState</code>	query · string	optional (default Active)	PreInstallation Staged Active

Response: `{ totalAssets, onlineCount, offlineCount, faultCount, powerFaultCount, flowFaultCount, lastUpdated, recentAlerts[] }`.

GET `/api/ewater/entities`

Entity hierarchy

Country → Organisation → Water System hierarchy with asset counts per water system.

POST `/api/ewater/proxy`

Generic eWater API proxy

Forwards an arbitrary request to any eWater upstream endpoint. For debugging and ad-hoc exploration.

Body: `{ "api": "state|query|command|auth", "path": "/api/...", "method": "GET|POST", "body": {...} }`

GET `/api/healthz`

Health check

Returns `{ "status": "ok" }`. No authentication required.

Overview & Authentication

The MCP server is exposed at `POST/GET/DELETE /api/mcp` using the **Streamable HTTP** transport (JSON-RPC 2.0). It provides the same data as the REST API as callable tools designed for LLM clients (Claude, GPT-4, WhatsApp AI agents, etc.). The server is **stateless** — a fresh server + transport is created per request.

Authentication

```
Authorization: Bearer <MCP_BEARER_TOKEN>
```

This is separate from the eWater credentials. Both must be configured: the bearer token protects the MCP endpoint itself; the eWater credentials (set via the REST API) are used by the tool implementations to reach live data.

Pagination Envelope

Every tool response uses the same envelope:

```
{
  "totalCount": 142,      // true total across all pages
  "returnedCount": 25,   // records in this response
  "offset": 0,
  "limit": 25,
  "hasMore": true,
  "items": [ ... ]      // or named array: "assets", "tagIds", etc.
}
```

Single-object tools (`get_asset_ewc_settings` , `get_asset_flow_rate` , `get_calibration_analysis`) return their result under `data` with `totalCount: 1` , `hasMore: false` .

Typical Call Chain

```
get_tag(nfcId)
→ get_asset(primary_asset_id)           // tap name, water system, country
→ get_household(household_id)          // household name
→ get_disbursements_by_tag_and_asset(nfcId, assetId) // water usage this month
→ get_asset_logs(assetId)               // recent device activity
```

Hierarchy Tools

`list_countries`

Lists all countries with organisation, water system, and asset counts. No inputs required. Always `hasMore: false` .

list_organisations

Lists organisations with water system and asset counts, optionally filtered by country.

Inputs:

Name	In / Type	Required	Description
countryId	integer	optional	Filter by country ID
countryName	string	optional	Filter by exact country name

list_water_systems

Lists water systems with asset counts. Pass the returned ID into `list_assets` or `list_registered_tags`.

Inputs:

Name	In / Type	Required	Description
organisationId	integer	optional	Filter by organisation ID
organisationName	string	optional	Filter by exact organisation name
countryId	integer	optional	Filter by country ID
countryName	string	optional	Filter by exact country name

list_assets

Paginated asset list. `totalCount` answers 'how many?' in a single call without paging everything.

Inputs:

Name	In / Type	Required	Description
limit	integer 1-100	optional (default 50)	Max assets per call
offset	integer	optional (default 0)	Pagination offset
status	string	optional	Active Staged PreInstallation Suspended
waterSystemId / waterSystemName	integer / string	optional	Scope to a water system
organisationId / organisationName	integer / string	optional	Scope to an organisation
countryId / countryName	integer / string	optional	Scope to a country

get_asset

Full details for one asset: name, type, status, GPS, water system, country. Always call this after `get_tag` to resolve asset details — never guess.

Inputs:

Name	In / Type	Required	Description
<code>assetId</code>	integer	required	From <code>list_assets</code> or <code>get_tag.primary_asset_id</code>

Asset Telemetry & Analysis Tools

get_asset_history

Time-series: tank height (water & chlorine), daily inflow, battery voltage, flow-rate history, dispense volume histogram. Each series independently paginated. For trend questions: 'was the tank stable last week?'

Inputs:

Name	In / Type	Required	Description
<code>assetId</code>	string	required	Numeric asset ID
<code>days</code>	integer 1-180	optional (default 7)	Days of history
<code>limit</code>	integer 1-2000	optional (default 500)	Max entries per time-series
<code>offset</code>	integer	optional (default 0)	Entries to skip per time-series

get_asset_ewc_settings

EWC device configuration: LCF (litres per tick — calibration factor), FCF (ticks per credit), preload charge, price per litre, currency/FX settings, and all other EWC device parameters.

Inputs:

Name	In / Type	Required	Description
<code>assetId</code>	string	required	Numeric asset ID

get_asset_flow_rate

Most recent dispense flow rate in L/min from the last 24 hours. `timedOut: true` means no dispense recorded — check for blockage or closure.

Inputs:

Name	In / Type	Required	Description
<code>assetId</code>	string	required	Numeric asset ID

get_calibration_analysis

NRW / meter-drift analysis: compares the configured LCF against measured dispense events, and compares preload against 'no credit' packet data. Returns LCF gap %, preload gap, and a plain-language interpretation. Use for NRW, meter drift, revenue loss, or calibration accuracy questions.

Inputs:

Name	In / Type	Required	Description
assetId	string	required	Numeric asset ID
days	integer 1-180	optional (default 30)	Days of dispense history to analyse

Tag & Household Tools

get_tag

Primary entry point for any tag lookup. Returns `primary_asset_id` , `household_id` , credit balance, disbursement/top-up counts, dates, and status flags (deleted, blacklisted). Always call this first when a tag ID is mentioned.

Inputs:

Name	In / Type	Required	Description
nfcId	string	required	8-character hex tag ID, e.g. DBDA4ED2 — case-insensitive

get_tag_info

Alias for `get_tag` — prefer `get_tag` for new calls.

Inputs:

Name	In / Type	Required	Description
nfcId	string	required	NFC tag ID

list_registered_tags

All fully-registered NFC tag IDs for a water system. `totalCount` = registered user fleet size — answers 'how many registered users?' from a single call.

Inputs:

Name	In / Type	Required	Description
waterSystemId	integer	required	From <code>list_water_systems</code>
limit	integer 1-500	optional (default 100)	Max tag IDs per call
offset	integer	optional (default 0)	Pagination offset

get_household

Household name and registration details. Never guess the name — always call this when you have a `household_id` . Phone number and address omitted per data-protection policy.

Inputs:

Name	In / Type	Required	Description
<code>householdId</code>	string (UUID)	required	From <code>get_tag.household_id</code>

get_household_info

Alias for `get_household` — prefer `get_household` for new calls.

Inputs:

Name	In / Type	Required	Description
<code>householdId</code>	string (UUID)	required	Household UUID

get_tag_usage

Individual dispense events for an NFC tag from the asset's EWC packet log. Fields: timestamp, litres (ticks ÷ LCF), creditConsumed, eventType (Dispense | No Credit). Use to verify household activity or debug credit deductions.

Inputs:

Name	In / Type	Required	Description
<code>nfcId</code>	string	required	NFC tag ID
<code>days</code>	integer 1-90	optional (default 30)	Days to scan
<code>offset</code>	integer	optional (default 0)	Pagination offset
<code>limit</code>	integer 1-200	optional (default 50)	Max events

Disbursement Tools

Draw from the eWater Usage API (daily aggregated totals). Each day bucket:

`estimateTotalLitres` , `totalTicks` , `totalSeconds` , `totalCredits` (null = no-credit), `readingCount` .

get_disbursements_by_tag_and_asset

Per-day aggregated water use for one tag at one specific tap. 'How much water did household X collect from tap Y this month?'

Inputs:

Name	In / Type	Required	Description
nfcId	string	required	NFC tag ID
assetId	string or integer	required	Asset ID of the dispenser
days	integer 1-365	optional (default 30)	Days to look back

get_disbursements_by_tag

Per-day totals for one tag across ALL taps. Multiple-tap usage on the same day is summed into one bucket. 'Total household consumption across all visits.'

Inputs:

Name	In / Type	Required	Description
nfcId	string	required	NFC tag ID
days	integer 1-365	optional (default 30)	Days to look back

get_disbursements_by_asset

Per-day totals from one tap across ALL tags. 'How much water was dispensed from tap X this month?' / 'How many people collected water here today?'

Inputs:

Name	In / Type	Required	Description
assetId	string or integer	required	Asset ID
days	integer 1-365	optional (default 30)	Days to look back

get_asset_logs

Fully-decoded device communication logs, newest first. **Critical:** the `pipeline` field identifies direction — `MQTT/UDP` = incoming telemetry from the device; `CmdApi` = outgoing command, NOT incoming data. If all visible entries are `CmdApi`, the device is offline. Decoded fields include `eventName`, `eventCategory`, `batteryVolts`, `flowTicks`, `litres`, `tagUid`, `vsen1-3` ADC (eSENSE), and more. Shengda NB-IoT frames include a plain-language `description` field.

Inputs:

Name	In / Type	Required	Description
<code>assetId</code>	string	required	Numeric asset ID
<code>days</code>	integer 1-30	optional (default 7)	Days of logs to fetch
<code>limit</code>	integer 1-100	optional (default 25)	Max entries — use 10-25 for WhatsApp
<code>before</code>	ISO timestamp	optional	Cursor for pagination — timestamp of last entry from previous call
<code>protocol</code>	string	optional	Include only: <code>Ewc2_5</code> <code>CommandApi_1</code> <code>4CCv1</code> <code>Gadwall</code>
<code>excludeProtocols</code>	string[]	optional	Protocols to exclude, e.g. [<code>"4CCv1"</code>]

Pagination Envelope

Used by all MCP list tools and the tag-usage REST endpoint:

Field	Type	Description
<code>totalCount</code>	integer	True total matching records across all pages — use this to answer count questions from a single call
<code>returnedCount</code>	integer	Records returned in this response
<code>offset</code>	integer	Records skipped before the first result
<code>limit</code>	integer	Max records per call as requested
<code>hasMore</code>	boolean	True when records exist beyond this page
<code>items / assets / tagIds...</code>	array	The records for this page, under a semantically named key

To iterate: increment `offset` by `limit` until `hasMore` is false.

Error Responses

HTTP Code	When	Body
<code>400</code>	Invalid request body or parameters	<code>{ "error": "..." }</code>
<code>401</code>	No credentials configured (REST) or invalid bearer token (MCP)	<code>{ "error": "..." }</code>
<code>404</code>	Asset / tag / household not found	<code>{ "error": "..." }</code>
<code>422</code>	Cannot compute result (e.g. no sensor data for range detection)	<code>{ "error": "..." }</code>
<code>502</code>	eWater upstream API returned an error	<code>{ "error": "eWater API error: ..." }</code>

Key Concepts

Term	Definition
LCF (LitresConversion)	Ticks per litre — the primary calibration factor. Divide raw flow-meter ticks by the LCF to get litres. Stored in EWC settings.
FCF (FlowConversion)	Ticks per credit (prepay unit). Different from LCF. The packet trailer bytes[33-34] hold the FCF, not the LCF — always read LCF from EWC settings.
Preload	Unmetered tick offset at valve-open — ticks dispensed before the meter starts counting. Must be subtracted when calculating true volume.
pipeline	Communication direction in packet logs: MQTT/UDP = device → platform (incoming telemetry); CmdApi = platform → device (outgoing command).
VSEN1/2/3	Analogue sensor ADC values (0-255) on eSENSE assets. ADC 51 ≈ 4 mA (sensor empty/off), ADC 255 ≈ 20 mA (sensor full). Convert to depth using sensor range from EWC settings.
NRW	Non-Revenue Water — water dispensed that does not generate revenue, typically due to meter drift, miscalibrated LCF, or preload offset errors.